

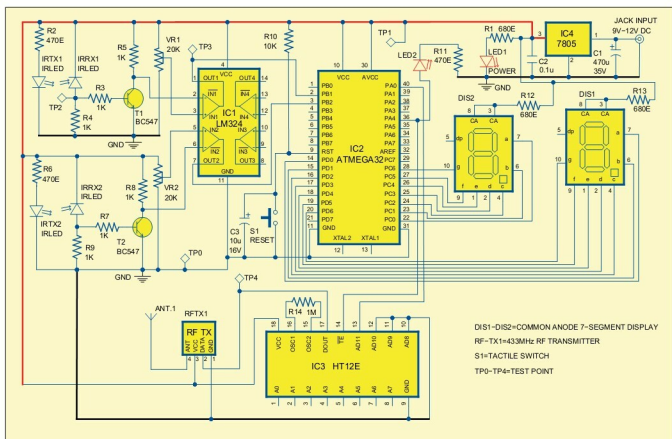


Fig. 2: Transmitter circuit

CMOS 8-bit microcontroller based on AVR enhanced RISC architecture.

It features 32 kB of in-system programmable flash program memory with read-while-write capabilities, 1024 bytes of EEPROM, 2kB SRAM, 32 general-purpose input/output (I/O) lines, 32 general-purpose working registers, three flexible timers/counters with compare modes, internal and external interrupts, a serial programmable USART, a byte-oriented two-wire serial interface, an 8-channel, 10-bit analogue-to-digital counter (ADC), a programmable watchdog timer with internal oscillator, an SPI serial port, and six software-selectable power-saving modes. In standby mode, the crystal/resonator oscillator is running while the rest of the device is in sleep mode. This allows very fast start-up at low power consumption.

Bidirectional counter. Infrared photodiodes IRTX1, IRRX1, IRTX2 and IRRX2 act as sensors for the bidirectional counter. Infrared LED trans-

Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	5V when IR LEDs on
TP3	5V
TP4	High pulse during transmission
TP5	High pulse during reception
TP6	High pulse when valid signal is received
TP7	5V when counter reaches zero
TP8	5V
TP9	12V
TP10	12V when dark/night
TP11	High for 13 minutes

mitter-receiver pair IRTX1-IRRX1 is to be installed at the entry point of the building. IRTX2-IRRX2 pair should be installed at the exit gate. The sensor output is amplified by transistor T1 and fed to LM324 op-amp (IC1), which, depending on the sensor output, gives a high or low signal to port pin PB1 (pin 2) of the microcontroller.

IRTX1 continuously transmits IR beam to the receiver, so the output of LM324 is high. When someone interrupts the IR beam, the output of IC1 goes low instantly and the program in the microcontroller increments the count by '1.' Similarly, for every interruption of the signal, the count increments by '1.' The number of counts is displayed on 7-segment displays DIS1 and DIS2, which are capable of showing up to 99 count.

Similarly, when someone interrupts the IR beam at the exit point, the output pin 7 of IC1 goes low instantly, which is fed to port pin PB2 of the microcontroller. The program in the microcontroller decrements the count by '1' for every interruption of the signal. When the count reaches '0,' port pin PA of the microcontroller goes high. Port pins PA0 and PA4 are connected to HT12E encoder IC (IC3). The output of IC3 is available at its data pin 17, which is fed to the RF-TX transmitter for transmitting the data wirelessly.